

Practice Problems 1

12/6/25 - 12/11/25

8) a) 12 ppl $\rightarrow$ split into 3 groups
1 team - 2 ppl
other teams - 5 ppl each

$$\binom{12}{2} \binom{10}{5} \frac{1}{2} = 66 \cdot \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \cdot \frac{1}{2} = 8316$$

b) $\binom{12}{4} \binom{8}{4} \frac{1}{3!} = 5775$

9) a)
$$\begin{array}{r} 111 \\ 110 \\ \hline 221 \end{array} \quad \begin{array}{r} 221! \\ \hline 111! 110! \end{array}$$

b) $\binom{221}{111} \times \binom{200}{100}$

15) $\sum_{k=0}^n \binom{n}{k} = 2^n$ $\binom{n}{k}$ specifies the # of subsets size k from a sample size n . The sum of the number of all subsets is equal to how many subsets from the sample of size n can exist. 2^n subsets exist because each element of the sample can either be in or out of a subset.

16) $n, k \quad n \geq k$
 $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$

2) Algebraic proof

$$\frac{n!}{k! (n-k)!} + \frac{n!}{(k-1)! (n-k+1)!} = \frac{(n-k+1)n! + k \cdot n!}{k! (n-k+1)!}$$

$$= \frac{(n+1)n!}{k!(n-k+1)!}$$

$$= \binom{n+1}{k}$$

b) Imagine you have n items and are choosing k of them. All of those combinations are valid when you choose k items from that sample + 1 more item. However all of those combinations exclude possibilities of k size that include that extra item. To calculate this gap, imagine that extra item is auto-included in the k -sized combination; now we need to find $k-1$ more items from the remaining n -sized sample, of which there are $\binom{n}{k-1}$ possibilities.

20) 2) $n \geq k \quad \binom{k}{k} + \binom{k+1}{k} + \dots + \binom{n}{k} = \binom{n+1}{k+1}$
Hockey stick identity

Imagine a sample of size $n+1$ where $n+1$ individuals

all are different ages. You select $k+1$ of them. Now, how many of those subsets are there such that the oldest person in the subset is...

the oldest in the sample? $\rightarrow \binom{n}{k}$
 " 2nd " " " " ? $\rightarrow \binom{n-1}{k}$
 $\vdots$
 " $(n-k)$ th " " " " ? $\rightarrow \binom{k}{k}$

b) $\binom{34}{4} + \binom{35}{4} + \dots + \binom{54}{4} - [\binom{4}{4} + \dots + \binom{33}{4}]$
 $= \binom{55}{5} - \binom{34}{5}$

24) $3! \cdot \underline{3!} - 1$

$$\frac{\quad}{6!} = \frac{\quad}{20}$$

$$25) \binom{11}{6} = \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7}{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4} = 462$$

$$\frac{461}{462}$$

$$28) \frac{1000 - 10 \cdot 9 \cdot 8}{1000} = 0.28$$

$$1 - \frac{6!}{66} \approx 0.9846$$

$$29) 2) >$$

$$b) \text{prob}(2 \text{ letter word is a palindrome}) \\ = \text{prob}(3 \text{ letter word is a palindrome}) = \frac{1}{26}$$

31) N elk n are captured & tagged
draw a sample of size m.

What is probability exactly k of m re-samples
are tagged?

$$\frac{\binom{n}{k} \binom{N-n}{m-k}}{\binom{N}{m}}$$

Ch 3 → hypergeometric!

33) 2) Choosing 2 balls, there's no difference in either
ball being green. Imagine you put two hands into
the bag & grab a ball in each hand, what you
call the first & second ball is no different either
way.

b) r red balls, g green balls

$$\text{prob first green: } \frac{g \cdot r + g(g-1)}{(g+r)(g+r-1)}$$

$$\text{prob second green: } \frac{r \cdot g + g(g-1)}{(g+r)(g+r-1)}$$

↔ equivalent

$$\begin{aligned}
 c) \quad g \cdot (g-1) + r(r-1) &= 2gr \\
 g^2 - g + r^2 - r &= 2gr \\
 g^2 - 2gr + r^2 &= g+r \\
 (g-r)^2 &= g+r = 16
 \end{aligned}$$

$$\begin{aligned}
 (g - (16-g))^2 &= 16 \\
 (2g - 16)^2 &= 16 \\
 2g - 16 &= \pm 4
 \end{aligned}$$

$$g = 6 \text{ or } 10$$

$$34) 2) \quad \frac{4 \left(\binom{13}{5} - 1 \right)}{\binom{52}{5}} \approx 0.199\%$$

$$b) \quad \frac{\frac{52.3}{2!} \cdot \frac{48.3}{2!} \cdot 44 \cdot \frac{1}{2!}}{\binom{52}{5}} \approx 4.75\%$$

answer key
has different
notation but
the math is the
same

42) probability no repeat word is all 26 letters

$$= \frac{26!}{\sum_{n=1}^{26} \binom{26}{n} n!} = \frac{1}{\sum_{n=1}^{26} \frac{1}{(26-n)!}} \approx \frac{1}{e}$$

$$48) \quad P_{\text{Arby}}(A \cup B) \neq P_{\text{Arby}}(A) + P_{\text{Arby}}(B)$$

$$P_{\text{Arby}}(A \cup B) = P_{\text{Arby}}(A) + P_{\text{Arby}}(B) - P_{\text{Arby}}(A \cap B)$$

↳ Arby is willing to pay

\$1000 $(P_{\text{Arby}}(A) + P_{\text{Arby}}(B))$ for a certificate

for event A and $\geq$ certificate for event B
 Arby is willing to sell a $A \cup B$ certificate
 for \$1000 ($P_{Arby}(A \cup B)$)

50) S: no spades C: no clubs
 H: no hearts D: no diamonds

$$\begin{aligned}
 & P(S) + P(C) + P(H) + P(D) \\
 & - P(S \cap C) - P(S \cap H) - P(S \cap D) - P(C \cap H) - P(C \cap D) - P(H \cap D) \\
 & + P(S \cap C \cap H) + P(S \cap C \cap D) + P(S \cap H \cap D) + P(C \cap H \cap D) \\
 & - P(S \cap C \cap H \cap D) \\
 & = \frac{4 \binom{39}{13}}{\binom{52}{13}} - \frac{6 \binom{26}{13}}{\binom{52}{13}} + \frac{4 \binom{13}{13}}{\binom{52}{13}} = 0.0511
 \end{aligned}$$

51) group of 7 prob days represent all seasons

$$\left(\binom{4}{1} 3^7 - \binom{4}{2} 2^7 + \binom{4}{3} 1^7 \right) \frac{1}{4^7} = 0.581$$

no 1 szn no 2 szns no 3 szns

$$1 - 0.581 = 0.419$$

54) classes meet 1/wk
 30 non-overlapping classes
 6 choices per day of week

student chooses 7 classes randomly. what is
 probability all 5 days represented?

$$\frac{5 \cdot \binom{6}{3} \binom{6}{1}^4 + \binom{5}{2} \binom{6}{2}^2 \binom{6}{1}^3}{\binom{30}{7}} \approx 0.3024$$

61) probability that the last passenger gets their assigned seat

There are 2^{99} ways to shuffle the seats.

Excluding the first seat, of the remaining, we can choose $0, 1, \dots, 99$ to be shuffled.

$$2^{99} = \sum_{n=0}^{99} \binom{99}{n}$$

We know it's correct to count each combination once because the seats selected to be shuffled can only be done in one way $\rightarrow$ are off and ascending.

For example, choosing 0 means the first passenger chose the right seat.

choosing 3 seats, X, Y, Z , we know the order has to be this one way (assume $X < Y < Z$ wlog)

$$\frac{X}{51} \dots \frac{Y}{5X} \dots \frac{Z}{5Y} \dots \frac{1}{5Z} \dots \frac{100}{100}$$

How many shufflings have 100? 2^{98}
b/c of the 98 other seats we can choose $0, 1, \dots, 98$ of them. $2^{98} = \sum_{n=0}^{98} \binom{98}{n}$

Therefore, the probability is $\frac{2^{98}}{2^{99}} = \frac{1}{2}$

The answer key's solution draws on similar observations but is more elegant. We notice

Seat 100 can only be filled by passengers 1 and 100. To all 98 other passengers, passengers 1 and 100 are symmetric. Therefore, $\frac{1}{2}$.